MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MODEL QUESTION PAPER - THEORY
STATISTICS

MG4DSCSTA200-Basics of Multivariate Distributions

Duration: 1.5 Hours **Maximum: 50 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer **any seven** questions. Each question carries **2 marks.**

1. Explain how bivariate data differs from univariate data, using a real-life example. [K][1]
2. Distinguish between Spearman's Rho and Kendall's Tau. [An][1]
3. Describe the difference between discrete and continuous bivariate distributions. Give one example of each. [U][2]
4. Define Marginal and conditional distributions of a bivariate distribution. when do you say that two random variables X and Y are independent [U][2]
5. Define non-singular multivariate normal distribution. [U][3]
6. Define quadratic form. [U][4]
7. If $X \sim N_P(0, I)$. Then the linear function B'X and quadratic form X'AX re independent iff AB=0. [A][4]
8. If $X \sim N_P(\mu, \sum)$, then show that $Q = N(\bar{X} - \mu)'\sum^{-1}(\bar{X} - \mu) \sim \chi^2_{(P)}$. [A][4]
9. What is the necessary and sufficient condition for X'AX to follow chi square distribution. [A][4]
10. State Cochran's theorem on quadratic forms. [U][4]

Part B

Short Essay Questions

Answer **any four** questions. Each question carries **6 marks**

11. Obtain the angle between two regression lines [A][1]

12. Ten competitors in a music competition were ranked by three judges in the following order:

Judge I	1	6	5	10	3	2	4	9	7	8
Judge II	3	5	8	4	7	10	2	1	6	9
Judge III	6	4	9	8	1	2	3	10	5	7

Which pair of judges have the nearest liking to the same type of music? [A][1]

13. (a) Define multinomial distribution. (b) Obtain the mgf of multinomial distribution. (c) Show that correlation coefficient between X_i and X_j is negative for all i and j.[A][3]

14. Derive the necessary and sufficient condition for the independence of sub vectors of a normal vector. [A][3]

15. If x~ N_P (0, I), find the necessary and sufficient conditions for independence of quadratic form X'AX and X'BX [A][4]

16. If x~ $N_P(\mu,I)$. then the quadratic form $(X-\mu)'\sum^{-1}(X-\mu)\sim \chi^2_{(r)}$, iff A is idempotent with rank r. [A][4]

Part C

Essay questions

Answer **any one** question. Each question carries **12 marks.**

17. (a)If $r_{12}=r_{13}=r_{23}=p$, show that $1-R^2_{1.23}=\dfrac{(1-p)(1+2p)}{1+p}$

(b) In a trivariate distribution: $\sigma_1=2$, $\sigma_2=\sigma_3=3$ $r_{12}=0.7$ $r_{23}=r_{31}=0.5$

Find (i) $r_{23.1}$ (ii) $R_{1.23}$ (iii) $b_{12.3}$ (iv) $b_{13.2}$ (v) $\sigma_{1.23}$ (vi) $\sigma_{1(23)}$ [An][1]

18. If (X,Y) follows BVN$(\mu_1,\mu_2,\sigma_1^2,\sigma_2^2,\rho)$, then prove that the marginal distributions of X and Y are also normal. [A][2]